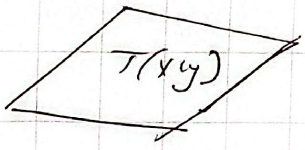
~~Lecture 7, 8, 9~~ Lecture 10, 11, 12, 13.

2D - Heat Diffusion + ~~Convection~~ ?



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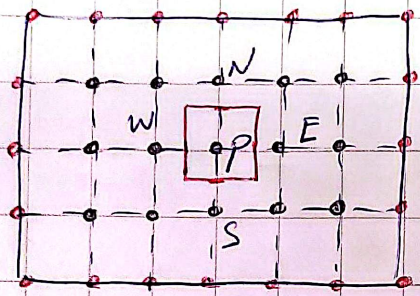
Source free : Pure diffusion "2D".

governing eqn : $\frac{\partial}{\partial x}\left(k \cdot \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k \cdot \frac{\partial T}{\partial y}\right) = 0$

$$\nabla \cdot (k \nabla T) = 0 .$$

nabla ←

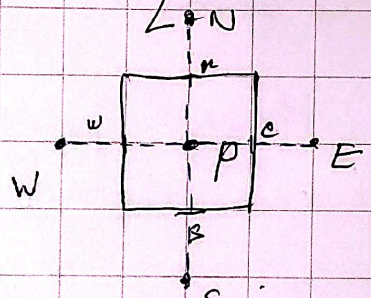
2D Grid + finite Volume.



2D grid.

L → Divide bandary length L into smaller lengths, both horizontal and vertically.

↳ grid (mesh) generation.



finite volume boundaries (faces) should lie midway between the grid points and have P as center.

Integrate governing eqn on finite volume:

Gauss is about converting a volume integral of a divergence.

$$\int \nabla \cdot (k \nabla T)\, dV = 0 .$$

$$\Downarrow \text{ gauss.}$$

$$\int \hat{n} \cdot (k \nabla T)\, dA = 0 .$$

$$\nabla T = \begin{pmatrix} \partial_x T \\ \partial_y T \end{pmatrix}$$

$\hat{n} : \hat{n}_e, \hat{n}_w, \hat{n}_n, \hat{n}_s$

Only 4 faces involved here, top and bottom surface not involved. No variation in these direction.

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$$\int \hat{n} \cdot (k \nabla T)\, dA = 0$$

$\hat{n}_s, \hat{n}_n, \hat{n}_e, \hat{n}_w$: $\begin{pmatrix}0\\-1\end{pmatrix} \begin{pmatrix}0\\1\end{pmatrix} \begin{pmatrix}1\\0\end{pmatrix} \begin{pmatrix}-1\\0\end{pmatrix}$ $\quad \nabla T : \begin{pmatrix}\partial_x T\\ \partial_y T\end{pmatrix}$

$$\int \hat{n}_e \cdot (k \nabla T)\, dA_e + \int \hat{n}_w \cdot (k \nabla T)\, dA_w$$

$$+ \int \hat{n}_s \cdot (k \nabla T)\, dA_s + \int \hat{n}_n \cdot (k \nabla T)\, dA_n = 0 .$$

$$kA\partial_x T\big|_e - kA\partial_x T\big|_w + kA\partial_y T\big|_n - kA\partial_y T\big|_s = 0 .$$

$$kA\partial_x T\big|_e - kA\partial_x T\big|_w + kA\partial_y T\big|_n - kA\partial_y T\big|_s = 0 .$$

diffusive flux across east face. : how much of Temperature is being transported by diffusion across the east face.

Decide how T varies between the grid points.

assume T varies piecewise linear. then.

Central differencing.

$$\partial_x T\big|_e = \frac{T_E - T_P}{h_{PE}} \quad , \quad \partial_x T\big|_w = \frac{T_P - T_W}{h_{WP}}$$

$$\partial_y T\big|_n = \frac{T_N - T_P}{h_{PN}} \quad , \quad \partial_y T\big|_s = \frac{T_P - T_S}{h_{SP}} .$$

$$k_e A_e \left(\frac{T_E - T_P}{h_{PE}}\right) - k_w A_w \left(\frac{T_P - T_W}{h_{WP}}\right) + k_n A_n \left(\frac{T_N - T_P}{h_{PN}}\right)$$

$$- k_s A_s \left(\frac{T_P - T_S}{h_{SP}}\right) = 0 .$$

$$k_e A_e \left(\frac{T_E - T_p}{h_{pE}}\right) - k_w A_w \left(\frac{T_p - T_w}{h_{wp}}\right) + k_n A_n \left(\frac{T_N - T_p}{h_{pN}}\right) - k_s A_s \left(\frac{T_p - T_s}{h_{sp}}\right) = 0$$

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$$\frac{k_e A_e}{h_{pE}} T_E - \frac{k_e A_e}{h_{pE}} T_p - \frac{k_w A_w}{h_{wp}} T_p + \frac{k_w A_w}{h_{wp}} T_w + \frac{k_n A_n}{h_{pN}} T_N - \frac{k_n A_n}{h_{pN}} T_p$$

$$- \frac{k_s A_s}{h_{sp}} T_p + \frac{k_s A_s}{h_{sp}} T_s = 0 .$$

$$- \left(\frac{k_e A_e}{h_{pE}} + \frac{k_w A_w}{h_{wp}} + \frac{k_n A_n}{h_{pN}} + \frac{k_s A_s}{h_{sp}}\right) T_p + \frac{k_e A_e}{h_{pE}} T_E$$

$$+ \frac{k_w A_w}{h_{wp}} T_w + \frac{k_n A_n}{h_{pN}} T_N + \frac{k_s A_s}{h_{sp}} T_s = 0 .$$

$$a_p T_p = a_E T_E + a_w T_w + a_N T_N + a_s T_s$$

with $a_p = a_E + a_w + a_N + a_s$.

a_p	a_E	a_w	a_N	a_s
$\frac{k_e A_e}{h_{pE}} + \frac{k_w A_w}{h_{wp}} + \ldots + \frac{k_n A_n}{h_{pN}} + \frac{k_s A_s}{h_{sp}}$	$\frac{k_e A_e}{h_{pE}}$	$\frac{k_w A_w}{h_{wp}}$	$\frac{k_n A_n}{h_{pN}}$	$\frac{k_s A_s}{h_{sp}}$.

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